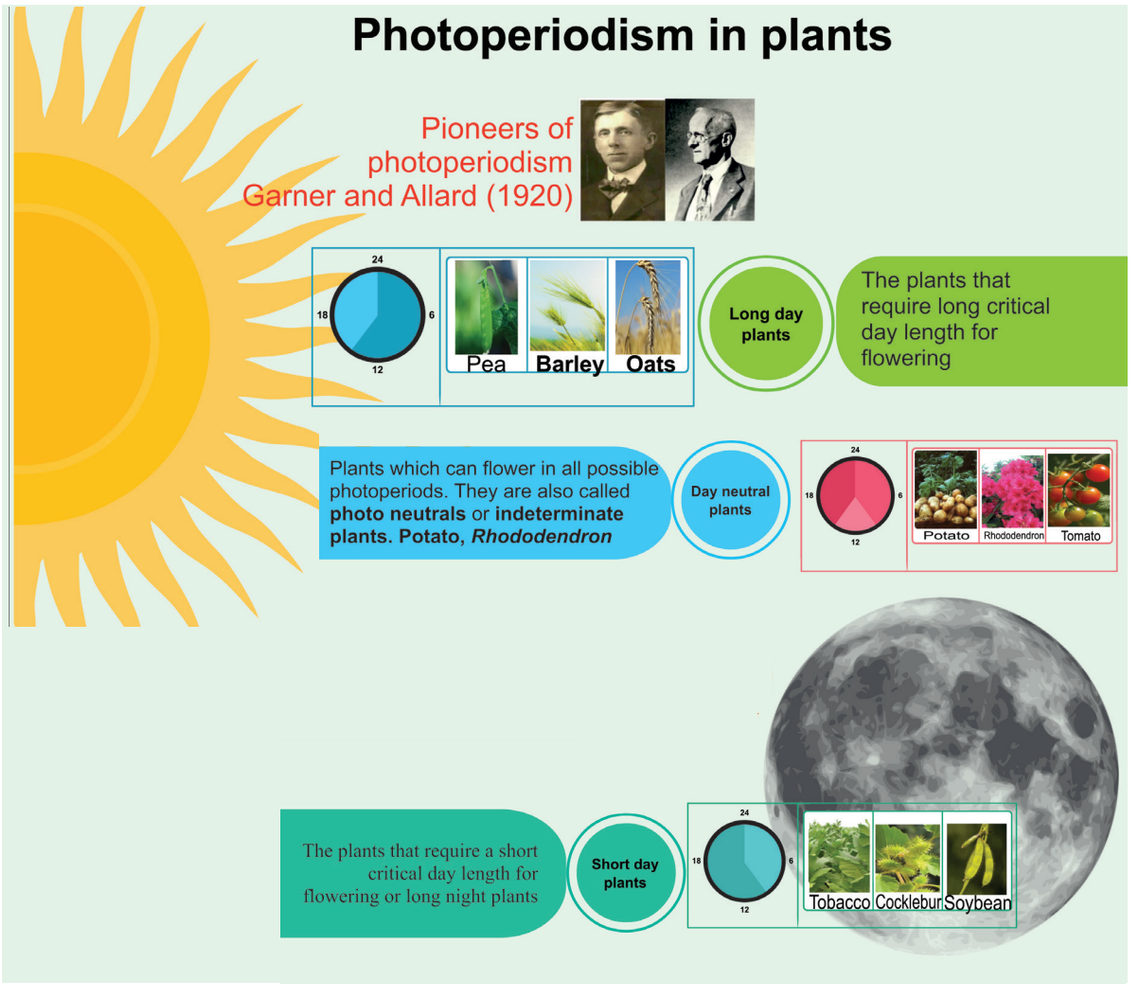


a role in seed germination and changes in membrane conformation.

15.5 Vernalization (*Vernal* – Spring Like)

Besides photoperiod certain plants require a low temperature exposure in their earlier stages for flowering. Many species of biennials and perennials are induced to flower by low temperature exposure (0°C to 5°C). This process is called **Vernalization.** The term Vernalization was first used by **T. D. Lysenko** (1938).

1. Mechanism of Vernalization:

Two main theories to explain the mechanism of vernalization are:

i. Hypothesis of phasic development
ii. Hypothesis of hormonal involvement

i. Hypothesis of phasic development

According to Lysenko, development of an annual seed plant consists of two phases. First phase is **thermostage**, which is vegetative phase requiring low temperature and suitable moisture. Next phase is **photo stage** which requires high temperature for synthesis of florigen (flowering hormone).

ii. Hypothesis of hormonal involvement

According to **Purvis** (1961), formation of a substance A from its precursor, is converted into B after chilling. The substance B is unstable. At suitable temperature B is converted into stable compound D called **Vernalin**. Vernalin is converted to F (Florigen). Florigen induces flower formation. At high temperature B is converted to C and devernalization occurs (Figure 15.21).

2. Technique of Vernalization:

The seeds are first soaked in water and allowed to germinate at 10° C to 12° C. Then seeds are transferred to low temperature (3°C to 5°C) from few days to 30 days. Germinated seeds

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